

Hui Guo

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Research Interests

My research is broadly in **statistical machine learning for imperfect data**. Methodologically, I am interested in learning with imperfect supervision, noisy label learning, weak and crowdsourced supervision, robust and distributionally robust learning, and uncertainty quantification. In an applied context, I am interested in developing models and algorithms for medical AI and foundation models for healthcare.

Education

- **Ph.D. in Computer Science, Western University** 09/2021–Present
Department of Computer Science
Advisors: Prof. Grace Y. Yi and Prof. Boyu Wang
Dissertation focus: **Learning with Imperfect Data: Statistical and Machine Learning Methods**
Expected completion: Summer 2026 | Graduate course average: 95.75%
- **B.Sc. in Mathematics and Applied Mathematics, Nankai University** 09/2016–06/2021
Thesis: A Generalization on Conditional Distortion Risk Measures
- **B.Econ. in Finance, Nankai University** 09/2016–06/2021
Thesis: Pairs Trading Strategy Based on Hurst Exponent
Undergraduate course average: 90.30%

Publications

Preprints and Submissions

- Dual-Granularity Learning for Regression with Continuous Noisy Labels
Hui Guo, Boyu Wang, Grace Y. Yi
Submitted to *Neural Information Processing Systems (NeurIPS)*, 2026.

Journal Publications

- Addressing both Variable Selection and Misclassified Responses with Parametric and Semiparametric Methods
Hui Guo, Grace Y. Yi, Boyu Wang
Bernoulli 32(2): 1303–1327, 2026. [\[paper\]](#) [\[code\]](#) [\[arXiv\]](#)
- Physician Effects in Critical Care: A Causal Inference Approach Through Propensity Weighting with Parametric and Super Learning Methods
Yuan Bian, Yu Shi, **Hui Guo**, Grace Y. Yi, Wenqing He
Journal of Data Science 23(1): 130–148, 2025. [\[paper\]](#)

Conference Publications

- Revisiting Source-Free Domain Adaptation: A New Perspective via Uncertainty Control
Gezheng Xu*, **Hui Guo***, Li Yi, Charles Ling, Boyu Wang, Grace Y. Yi (*Equal contribution)
International Conference on Learning Representations (ICLR), 2025. [\[paper\]](#) [\[code\]](#)
- Learning from Noisy Labels via Conditional Distributionally Robust Optimization
Hui Guo, Grace Y. Yi, Boyu Wang
Neural Information Processing Systems (NeurIPS), 2024. [\[paper\]](#) [\[code\]](#) [\[arXiv\]](#)
- Label Correction of Crowdsourced Noisy Annotations with an Instance-Dependent Noise Transition Model
Hui Guo, Boyu Wang, Grace Y. Yi
Neural Information Processing Systems (NeurIPS), 2023. [\[paper\]](#) [\[code\]](#)

Working Papers

- Instance-Dependent Bayesian Modeling and Robust Training with Crowdsourced Noisy Data
Hui Guo, Grace Y. Yi
Manuscript in preparation.
- Reliability-Aware Longitudinal Clinical Prediction from Multimodal Electronic Health Records
Hui Guo, Yi Wan, Boyu Wang, Grace Y. Yi
Manuscript in preparation.

Awards and Distinctions

- **Robert and Ruth Lumsden Graduate Fellowship in Science**, Western University 2024–2025
Awarded to one PhD student per department in Science, selected based on academic excellence, research ability, and leadership, with criteria aligned with NSERC doctoral scholarship standards.
- **The Fourth Research Challenge Contest of Nankai University** May 2019
Awarded honorable prize.
- **Students' Innovation and Entrepreneurship Training Program** 2018–2019
Team was awarded the excellence award.
- **Nankai University Academic Excellence Scholarship** 2017–2018
Awarded to the top 10% of students based on academic performance.
- **National Mathematical Modeling Contest for University Students** September 2017
Team was awarded second prize in Tianjin.
- **Nankai University Talent Scholarship** 2016–2017
Awarded to the top 10% of students based on academic performance.

Selected Talks

- Learning with Noisy Labels: Variable Selection and Misclassification Probability Modeling
Lightning talk for Dr. Raymond Carroll's visit, Western University, June 2025.
- Revisiting Source-Free Domain Adaptation: A New Perspective via Uncertainty Control
International Conference on Learning Representations (ICLR), Singapore, April 2025.
- Learning from Noisy Labels via Conditional Distributionally Robust Optimization
Neural Information Processing Systems (NeurIPS), Vancouver, Canada, December 2024.
- Label Correction of Crowdsourced Noisy Annotations with an Instance-Dependent Noise Transition Model
Neural Information Processing Systems (NeurIPS), New Orleans, USA, December 2023.
- Variable Selection for Logistic Regression Models with Misclassified Response
Annual Meeting of the Statistical Society of Canada, Online, May 2022.
- Personalized Physician Recommendation for Critical Care Using the TreeSHAP Method (case study)
Annual Meeting of the Statistical Society of Canada, Online, May 2022.

Workshops & Events

- Remarkable, Vector Institute
Toronto, ON, February 2026
- Southern Ontario Learning Theory (SOLT) Workshop, Vector Institute
University of Waterloo, Waterloo, ON, November 2025

Academic Service

- **Conference reviewer:** ICML 2026; ICLR 2024–2026 (3 years); AISTATS 2024–2026 (3 years); NeurIPS 2024–2026 (3 years).
- **Journal reviewer:** *Foundations and Trends in Signal Processing*; *Neural Networks*.

Selected Teaching Experience

Teaching Assistant, Western University

London, ON, Canada

- **CS2209: Applied Logic for Computer Science** 2021–2026 (7 terms)
An undergraduate core course for Computer Science students, with around 300 students, covering propositional and predicate logic, resolution, and logic programming in Prolog.
- **CS2035: Data Analysis and Visualization** 2022–2024 (3 terms)
An undergraduate elective for Computer Science students, with around 80 students, covering statistical data analysis and visualization techniques using MATLAB.

Technical Skills

- **Programming:** Python, R, MATLAB, C++, $\text{\LaTeX}$, Prolog.
- **Languages:** English (fluent), Mandarin Chinese (native).